

Eyan Documet

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SUMMARY

First-principles mechanical engineer driven by the exploration and execution of novel, complex, and coupled electromechanical and multiphysical systems. Proven history of executing high-level design goals across the domains of precision mechatronic positioners, custom simulation suites, and first-of-kind multiphysics assemblies.

SKILLS

- **Engineering & Analysis:** Geometric Dimensioning and Tolerancing (GD&T), Tolerance Stack-Up Analysis, Computer Aided Engineering (CAE), Finite Element Analysis (FEA), Discrete Element Analysis (DEA), Design For Manufacturing/Assembly/Testing (DFx), Design of Experiments (DoE), Mechanisms, Manipulators
- **Mechatronics:** Motion/Trajectory planning, MuJoCo, linear circuit design, control theory (PID, LQR), simulation, motor sizing, Microcontrollers (Arduino, ESP32), Programmable Logic Controllers (PLCs)
- **Software:** PTC Creo, Windchill PLM, Autodesk Inventor, Autodesk Fusion, Python (NumPy/SciPy), MATLAB, C/C++ (Arduino, PlatformIO), Excel, Linux, Git, Codeware COMPRESS, ANSYS

EDUCATION

University of California, Berkeley, B.S. Mechanical Engineering Dec 2025

- **Selected Courses:** Robotic Locomotion, Finite Element Methods, Simulation for Advanced Manufacturing

EXPERIENCE

Mechanical Engineer I, ASC Process Systems - Valencia, CA May 2026 – Present

- Owned mechanical engineering and execution of two 500+ component, industrial-scale autoclave systems for aerospace clients, delivering CAD, BOMs, fabrication drawings, and engineering release packages
- Designed and released custom assemblies including water-cooling jackets, rail systems, and mechanical safety interlocks while coordinating cross-functionally with manufacturing, purchasing, and production teams
- Led root-cause investigations into recurring fabrication non-conformances, updating drawings and engineering standards to improve manufacturing robustness and reduce shop-floor rework.

R&D Engineering Intern, Lawrence Berkeley National Lab - Berkeley, CA Jun 2025 - Dec 2025

- Independently developed a first-of-kind, high-precision magnetic positioner assembly for next generation synchrotrons, promising $\sim 10\times$ reduction in mass/power requirements versus conventional alternatives
- Programmed an automated 2D magnetics simulation suite in MATLAB, reducing design iteration time from 1-2 hours per assembly to near-instantaneous by leveraging inventory-level constraints and custom field simulations
- Resolved 50+ design conflicts in PTC Creo and Windchill PLM, focusing on manufacturing scalability and Tolerance Stack-up Analysis, while designing and releasing remaining components for CNC and MJF printing
- Created and deployed a custom PID-based brushed motor controller (achieving sub-degree precision) using Arduino microcontrollers and Hall-effect encoders to allow for robotic positioning of in-situ magnetic assemblies

Course Reader, ME154/254: Thermophysics, UC Berkeley - Berkeley, CA Aug 2025 - Dec 2025

- Provided technical feedback and evaluation for a cross-listed cohort of graduate and undergraduate students.

FEATURED PROJECTS

3D-Printed SCARA Manipulator, Solo Lead Jul 2025 - Present

- Developing a stepper-driven SCARA-style robotic arm from first-principles using Python and Fusion 360
- Created scripts for motor sizing, forward/inverse kinematics, D-H parameters, and dynamic equations of motion

Gyroid Optimization Experiment, Project Lead Aug 2025 – Dec 2025

- Led a 3x3 full-factorial experiment to identify the existence of an energy absorption optimum in complex structural geometry, proving so with a coefficient of determination $R^2 = 0.94$.

Autonomous Firefighting Robot, Mechatronics Lead Feb 2025 – Jun 2025

- Designed and built a fully-autonomous fire-prevention robot featuring 3D-printed electronics enclosures, inline fluid delivery, and a custom low-level computer vision driver built for the ESP32 microcontroller